

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6.	(a)	(i)		Quaternary	1			1		
		(ii)		Translation of mRNA by ribosomes/ production of {polypeptides/ protein synthesis} on ribosomes (1) (Packaging of polypeptide) into vesicles/ transport to Golgi (body) (1)		2		2		
		(iii)		Any two (x1) from • {Assembly/ joining} of {polypeptides/ subunits/ protein} (into quaternary structure) (1) • addition of carbohydrate/ glycosylation/ produce glycoprotein/ owtte (1) • Packaged (into vesicles) (1)		2		2		
	(b)			• Fibrinogen (is globular, so it) dissolves in plasma (1) • When it is converted to Fibrin it {becomes insoluble/ forms fibres} (1) • (fibres cross a wound and) {seal it/ trap blood cells/ owtte} (1)		2	1	3		
	(c)			Peptide bond (1) The active site of thrombin is (only) complimentary to {the arginine-glycine bond/ the enzyme}/ ORA thrombin is specific to the arginine - glycine bond (1)	1	1		2		

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	(d)	(i)		A. {Hirudin/ it} acts as a competitive inhibitor (1) B. {Fits into/ binds to/ complimentary to/ competes for} active site (on thrombin) (1) C. {Preventing/ reducing} {ESCs/ or description of} (1) D. (Hirudin slows down the action of Thrombin) so that blood {clots more slowly/ does not clot} (1)	2	1	1	4		
		(ii)		Used to {prevent/ reduce risk of/ slow down formation of} blood clots/ DVT/ anticoagulant/ owtte			1	1		
				Question 6 total	4	8	3	15		